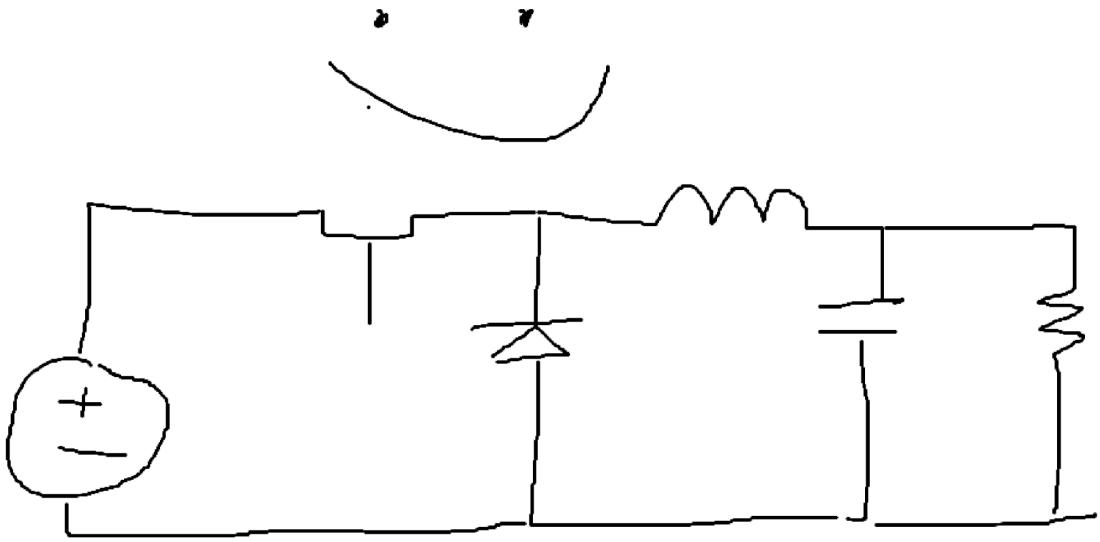
Formula Slug – Buck Converter Schematic Block Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

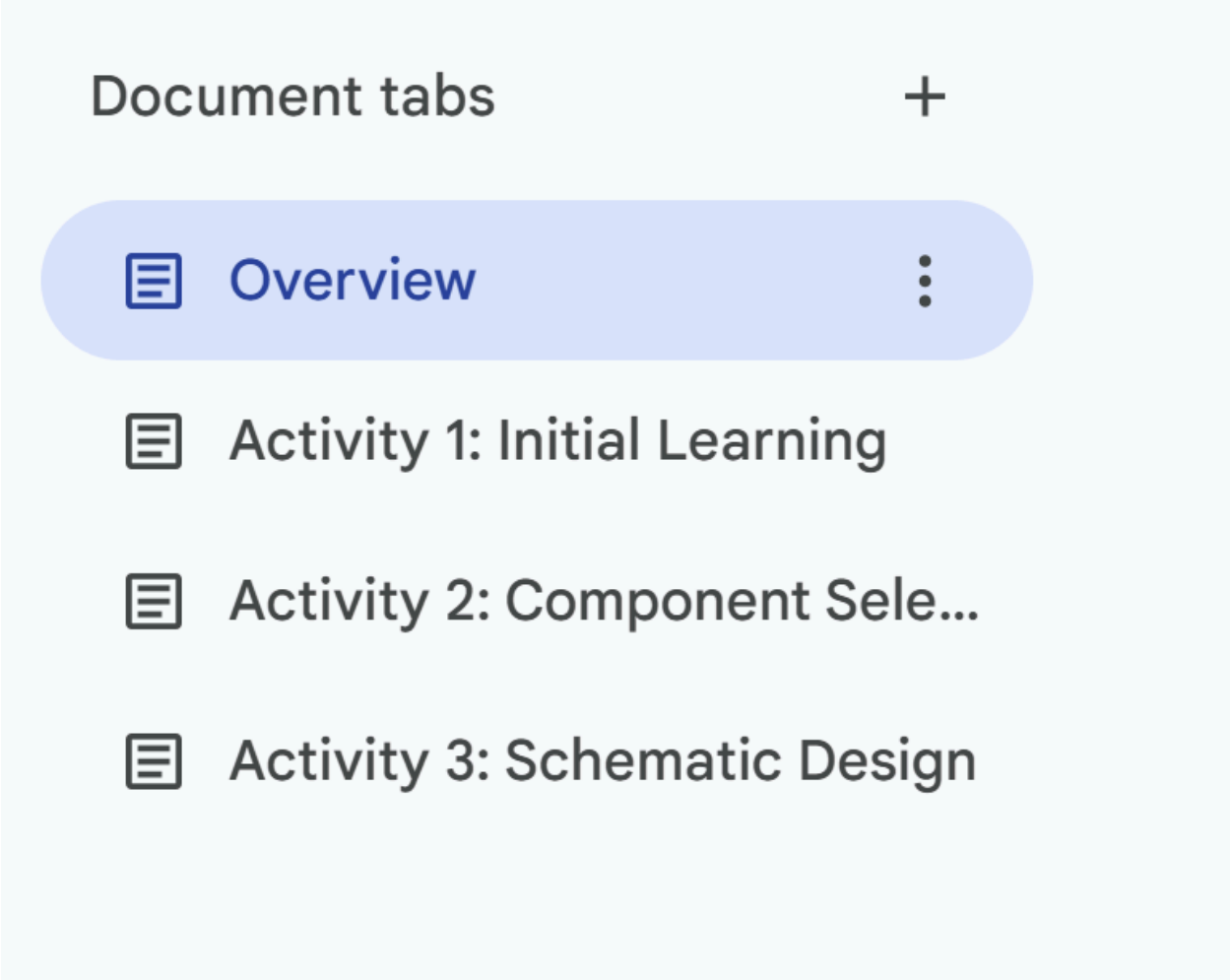
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
   https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
3. Khan Academy, Compound Circuits:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power
4. EEVBlog, Voltage Dividers:
   https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
   https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
   a. Lecture 5: Intro to DC/DC, Part 1
2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

OK OK LETS GOOOOOOOOOOO PE ONBOARDING

Ohm's law video:

- I already know most of this but its a good review
- Ohms law: V= IR where I is current in amps and R is resistance in ohms
- As voltage in circuit increases the current will increase provided the resistance stays the same
- If the resistance goes up the current will go down if the voltage is held constant
- Essentially Voltage and current are proportional to each other and resistance ansd current are inversely related to each other

Ex

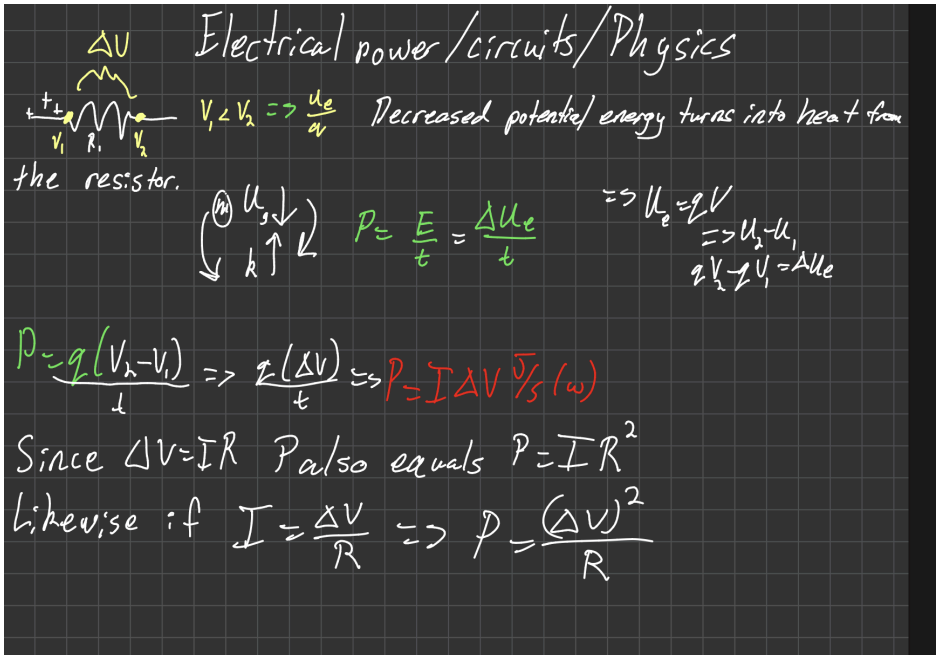
- 12 V battery connected across resistor (4ohm). What is current? 12/4 or 3 given by ohms law equation

EX

- 3 resistors connected in series (R1+R2+R3) where r1 = 3 and r2=4 and r=5 and 60 v → 60/12 = 5 current
- To find the v across a resistor we use ohms law w.r.t a single resistor. Current is consistent through all resistors but resistance only holds true to one resistor
- Kierchoff's law: as you go around a circuit in a loop the total voltage will be 0. Id sketch this out normally but its pretty easy to picture.
- In a parallel circuit rtotal = 1/r1 +1/r2 … Current total = I1+I2 …

Khan Academy, Compound Circuits

- PDF notes submission



The rest of the notes are more mathematical computations to calculate what I will actually need for the BUCK and Ill throw those into activity 2

BuActivity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



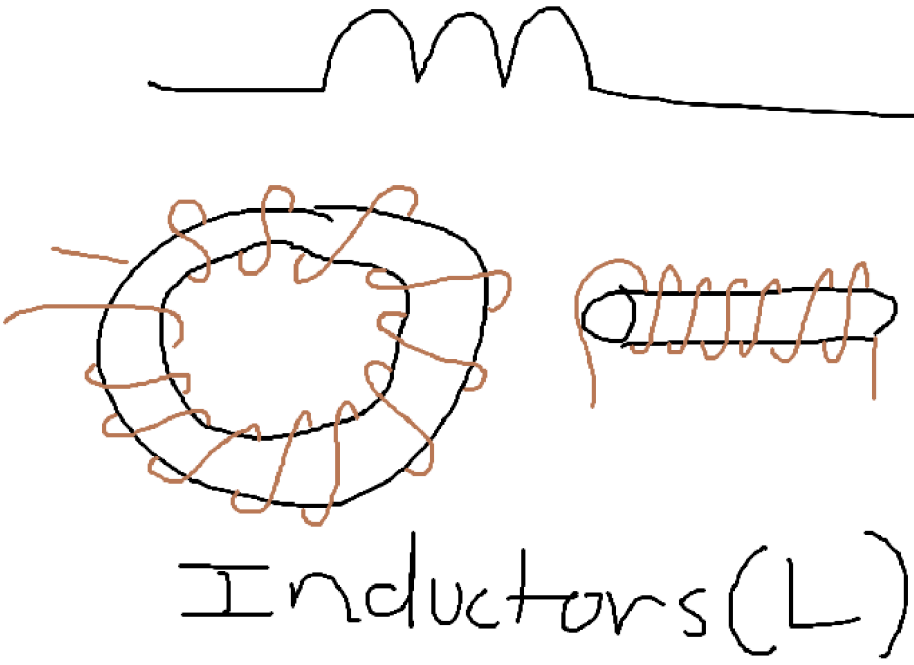
The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

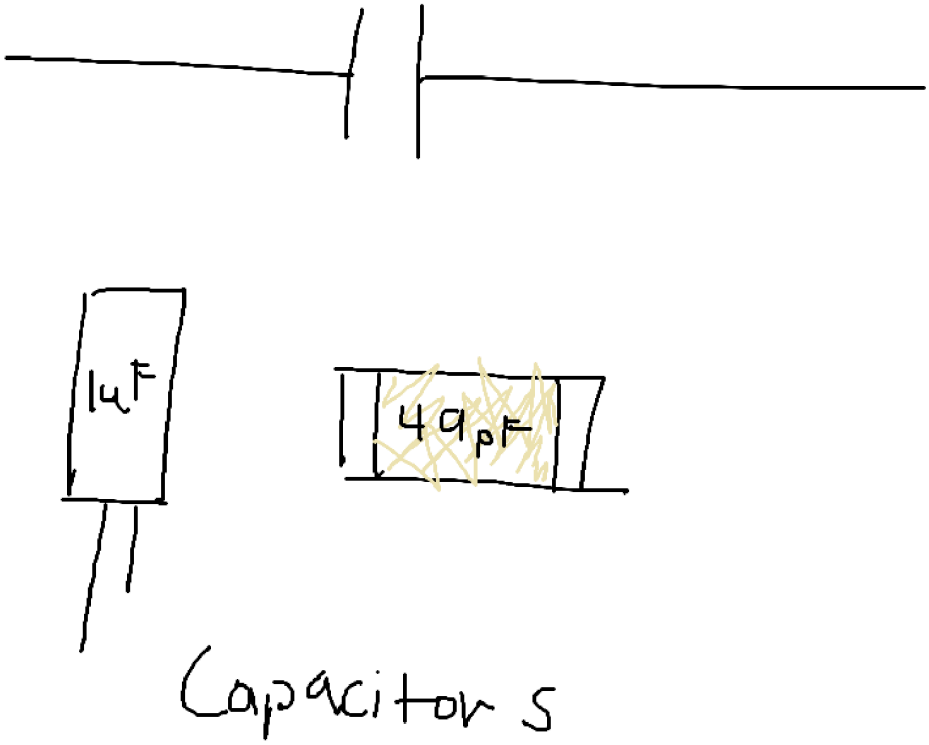
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

Component	Cost	Link	Units	Size
Buck IC: TI TPS54202DDCR	$1.63	https://www.digikey.com/en/products/detail/texas-instruments/TPS54202DDCR/6005348	1	2.9x1.6mm
Inductor: Bourns SRP6540-100M	$1.16	https://www.digikey.com/en/products/detail/bourns-inc/SRP6540-100M/3986099	1	(7.2x6.5x4mm)
Input Cap: Samsung CL31B106KBHNNNE	$0.42	https://www.digikey.com/en/products/detail/samsung-electro-mechanics/CL31B106KBHNNNE/5961251?s=N4IgT	1	(3.2mm x 1.6mm)

		CBcDaIMIBkDMBGAQigDANgNJoAkA5EgURAF0BfIA		
Output Caps: Samsung CL21A226MOQNNNE	$0.96	https://www.digikey.com/en/products/detail/samsung-electro-mechanics/CL21A226MOQNNNE/3888567?s=N4IgTCBcDaIMIBkwEYCCYwDYCyB5AigHLECiIAugL5A	3	(2.00mm x 1.25mm) x3
Bypass/Bootcaps: Samsung CL10B104KB8NNNC	$0.30	https://www.digikey.com/en/products/detail/samsung-electro-mechanics/CL10B104KB8NNNC/3886658	3	(1.60mm x 0.80mm)
Feed Forward cap: Murata GRM1885C1H680JA01D	$0.13	https://www.digikey.com/en/products/detail/murata-electronics/GRM1885C1H680JA01D/586916?s=N4IgTCBcDaIOICUCyBGAHGgrAYRQCQDY0AGAKQEFiUAREAXQF8g	1	(1.60mm x 0.80mm)
Resistor top: YAGEO RC0603FR-07100KL	$0.10	https://www.digikey.com/en/products/detail/yageo/RC0603FR-07100KL/726889?s	1	(1.60mm x 0.80mm)

		=N4IgTCBcDaIEoGEAMA2JBmAYnAtEg7AIxJIDSAMiALoC%2BQA		
Resistor bottom: YAGEO	$0.10	https://www.digikey.com/en/products/detail/yageo/RC0603FR-0713K3L/726931	1	(1.60mm x 0.80mm)
Total Cost	$4.80			

Note* yo edmund i feel like its important to note basically every single part but my inductor is out of stock but they are re stocking like every part in either late oct or early nov just to lyk, didnt want u to think it was an oversight or anything
Also id look at this page when u go thru the kicad schem cause its also my notes for learning how to make a buck converter aswell as explaining design choices
THE "what it does" is my notes cause its pretty new to me
Why TI TPS54202DDCR? I think i referenced this as the variable U1 in my schem
What it does: Brain and switches, contains both mosfets, control loop, and protection features
Minimums:

- Input of 10-15V
- 2A output
- Switching frequency of 500khz minimum
- Synchronous: hiccup-mode overcurrent, and undervoltage lockout so it won't run below about 4.2V input.
- SOT-23 package: leads you can see, probe and hand-solder, easily modifiable
- Our IC meets all these requirements while being pretty cheap :)))))))

Why Bourns SRP6540-100M: this is L1
Stores energy each cycle adn smoothes the **choppy asf** switch voltage into a steady current
Minimums

- Keeps ripple at 20-35% of our current load
- Doesnt saturate (margin on .2 so not ideal)
- **Low resistance so we dont overheat and absolutely fry all of our hard work**
- Once again cheap cheap

(**Edmund** idk if you will see this but ima be real with you I used AI to determine if the **SRP6540-100M** actually met the requirements I wanted. It solved for ripple current, peak

current and saturation, heating, and the output voltage and ripple cause i lwk had absolutely no clue how to calculate this, CSE 12 and Physics 5C is my only related knowledge aside from my own work)

Why Samsung CL31B106KBHNNNE input cap: this is c1

What it does: The input current arrives at as pulses (2A, then oA every 2mew(s)). This capacitor supplies those pulses locally so that the wiring harness doesnt carry a high frequency current

Minimums:

- Our buck IC needs at least 10 mew F of ceramic capacitance
- It has to handle the ripple current (worst case is .5)
- Also car buses have a lot of voltage spikes because of the extremities experienced in the hardware so it makes more sense for a chunkier 50v capacitor. Its also has sa high capacitance ceiling, just seemed like a good build.

Why Samsung CL10B104KB8NNNC bypass/bootcaps this is c2

What it does: handles the hella fast edges of current pulses

- A large capacitor has internal inductance which essentially makes it hella slow to respond to fast changes **(Bear vs Panther, in this case we want the panther)**
- **I lwk found this part via research,** apparently its really common and reliable, cheap and always in stock/stocking
- 50V gives the same margin as C1 (consitency)

Why Samsung CL10B104KB8NNNC same as c2: labeled as c3

Same part, different function, (bootstrap)

What it does:

- This time the **CL10B104KB8NNNC** powers the high of the mosfets gate driver
- C3 charges to 5v while SW is low
- When the sw rises to 12, c3 lifts boot to 17 i think
- Also this part is the same as the previous and the same as c2 and c7 so it makes assembly way easier, dont gotta assemble 234934 different parts

Why Samsung CL21A226MOQNNNE (all our output caps) = c4-c6

What it does: it absorbs the inductors ripple current so that our output stays flat and consistent(exactlyt what we need or else ts wont work)

Also supplies sudden load changes while the control loop catches up

Minimums: (USED AI FOR THIS CALCULATION EDMUND DONT HATE ME)

- Ripple of 1% only needs about 3.3mew F acording to C_out > deltaI_L/8_fsw x deltaV
- TI's design calculation for the response needs like 25 mew F which is the number that determines the size

Why 3 of em?

- Well with AI's calculation and there being a DC bias we would need 3 of 10 mew F to get over our 25 mew F limit. The ripple then decreases massively as well.

Why CL10B104KB8NNNC (c7):
What it does: inverse of c2 (output version of c2), catches the high frequency switching spikes that the 22mewf caps are too slow to catch
OUR RESISTORSSSS (YAGEO RC0603FR-07100KL
What it does: Scales the 5v down to the chips reference V, chip adjusts its switching until it hits target
From ohms law (ai calc):
V_out = .596(1+r1/r2)
V_out = 0.596(1+100/13.3)
V_out = 5.08 V
And from what I understand Yageo is always in stock/stocking and is reliable and consistent
Why Feed Forward cap: Murata GRM1885C1H680JA01D (c8):
What it does: sits across r1 and adds 0 to the control loop, gives a phase boost giving better stability and speed/effeciency

Our EN pin:

- Internal .7 mew a turns chip on when when the pin is left floating
- The ABSOLUTE max is 7V so tying EN to the 12 bus (I highlighted this in the schematic) will fry our shit
- No costs and avoids the failure mode

Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.

Yo wsp edmund idk why but I put everything that kinda references the schematic and a lot more schematic notes in activity 2 so id go read that before or in tandem with looking at the kicad schematic

I went to an office hours but forgot to ask about the PCB which im sure is apart of the schematic, i mean yes the schematic is the main part but the PCB brings it to life.
PCB Design layout:

- U1 is in the middle The SW connection runs straight down into L1 ona short, wide trade so the loop stays small (efficient)
- C3 sits right next to U1's boot pin, and its SW side ties in just below the chip
- The 12 V input comes in at J1, and runs along one wide trade thru c1 and c2 into the vin pin
- 5V output: L1 feeds the 5v rail at bottom, that runs thru c4-c7 to J2 (very similar to schematic)
- Feedback network (resistor work): R1, r2, and c8 sits next to U1's FB pin away from L1. it measures 5v right at the output caps w the trace on the bottom layer so its far from the noisy sw node
- Ground: both layers are filled w ground copper, connected by vias ground pads
- Traces: power traces should be around 1mm and ,3-.4mm for signal traces (from notes)

EDMUNDDDD
I know how the PCB layout should look but I don't have enough time to actually spend the hours I would need to to sketch this out in kicad.
Therefore I briefly collaborated w AI by giving them the above notes and had them sketch out the PCB that will be turned in and committed into the repo(i will title it so you know its not

100% my work). IK and i assume you do as well that people will use AI on this project and turn in AI slop but I want to be clear about what is mine and what isn't.

Extra Activity: Tractive System Shunt Current and/or Voltage Sense

If you happen to finish the buck converter, or you feel that the buck converter doesn't speak to you, you can also work on this project as well. This one will be in the form of a presentation / research project.

This project is based around adding to our current sensors within our Tractive Battery's container. Currently, we use a hall effect current sensor, which kind of looks like this:



For accuracy and data acquisition reasons, it would be nice to be able to add to this hall effect sensor with a shunt one, where you measure the voltage drop across a specially manufactured piece of metal with a specific resistance and voltage drop across, where you can then determine what current is going through it based on it. However, there are some constraints to this system, ie: isolation, heat dissipation, etc

Our current tractive battery current can go up to ~600A peak. We want to be able to measure up to this and down to the negatives as well in order to track regen current. Required by FSAE 26-27 Rules (EV.8.7.4), we need to be able to *prove* to judges that the Brake System Plausibility Device works through simulating tractive system current going through the current sensor. Find a way to be able to simulate this using a shunt if possible. At the moment, we have wire wrapped around a hall effect sensor multiple times, simulating higher currents with a smaller test current.

We would also like to be able to properly measure the voltage in our tractive system while we drive. We should find a way to take that voltage measurement, isolate it and pass it through to a

microcontroller so that we can have this data for the future.

When completing this research project, continue to write good documentation and make sure to optimize for costs when possible.

Have fun!